

9.3 Reactor

Reactor animates objects automatically to simulate the effect of real-world forces and collisions. Reactor can create rigid bodies, which do not deform as they collide, or soft bodies, which change shape as they collide and move. Reactor creates highly realistic looking animation and is good for simulating natural motion. Reactor has its own menu on the main menu bar, which contains all of reactor's objects and modifiers, as well as tools to define object properties and to create and preview animations. Reactor objects can also be found in the Create panel, under Helpers. The Utilities panel also has a Reactor button that allows you to configure the dynamics engine and create animations.

9.3.1 Rigid Body Dynamics

Reactor can create rigid body dynamics to simulate how physical objects move in the real world. These tools can be used to simulate objects falling, colliding, or being affected by physical forces such as wind and gravity. The solutions can be physically accurate, making the result very realistic.

9.3.2 Creating a Simulation

Rigid body simulations are set up by creating a rigid body collection containing each object in the simulation, defining each object's physical properties, adding in forces, and then running the simulation.

Create a rigid body collection (reactor→Create Object→Rigid Body Collection). There is a list containing all the objects in the simulation. Objects can be picked individually from a viewport or added from the list. Each object has a set of object properties that define physical properties about the object such as mass, friction, and elasticity. These are accessed via reactor→Open Property Editor. Objects that remain passive, such as the walls and the floor, are given a mass of zero, while active objects, such as the ball and boxes, are given masses larger than zero. After the properties are set up, the animation can be created. The reactor tab of the Utilities panel contains tools to create and preview the animation. The tab also contains a rollout for configuring the parameters of the world, such as the force of gravity. Animation previews appear in the Real-Time Preview window. When the preview is approved, the final animation can be generated. This creates keyframes for all the objects in the scene.

9.3.3 Forces

In addition to reactor's built-in gravity, other forces can be used to create